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> -----
      name: <unnamed>
      log:  F:\ECON 408\Practice do file\Homework-3.log
      log type: text
      opened on: 16 May 2026, 12:56:46

.
. clear all

. set seed 1000

.
. use "HOMEWORK 3.dta" , clear

.
. rename t time

. tsset time

Time variable: time, 1 to 100
      Delta: 1 unit

.
.
. *
. *
. * Identification
. *
. * (1) Time Series plot of variable y
. * //-----
.
.
. set scheme white_tableau

.
. tsline y, ///
>         lcolor(navy) lwidth(medthick) ///
>         title("{bf:Time Series Plot of y}", size(large)) ///
>         xtitle("Time Period", size(medlarge)) ///
>         ytitle("Values", size(medlarge)) ///
>         ylabel(, labsize(medium)) ///
>         tlabel(, labsize(medium)) ///
>         graphregion(color(white)) xsize(8) ysize(4) ///
>         name(tsplot_hw3, replace)

.
. graph export tsplot_hw3.png, name(tsplot_hw3) replace
file tsplot_hw3.png saved as PNG format

.
.
. * (2) Stationarity Test
. * //-----
.
.
. varsoc y , maxlag(12)

Lag-order selection criteria

```

Sample: 13 thru 100							Number of obs = 88		
Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC	
0	-196.952				5.2647	4.4989	4.51024	4.52705	
1	-126.231	141.44	1	0.000	1.07947	2.91434	2.93702	2.97064	
2	-122.116	8.2302	1	0.004	1.00571	2.84354	2.87757	2.928	
3	-122.086	.0607	1	0.805	1.02815	2.86558	2.91095	2.97819	
4	-111.977	20.216*	1	0.000	.835959*	2.65858*	2.71529*	2.79934*	
5	-111.842	.27143	1	0.602	.852618	2.67822	2.74627	2.84713	
6	-111.302	1.0795	1	0.299	.861691	2.68868	2.76807	2.88574	
7	-110.755	1.0948	1	0.295	.870745	2.69897	2.7897	2.92418	
8	-110.701	.10755	1	0.743	.889864	2.72047	2.82255	2.97384	

```

| 9 | -110.575 .2522 1 0.616 .907958 2.74033 2.85375 3.02185 |
| 10 | -110.484 .18236 1 0.669 .927211 2.76099 2.88575 3.07066 |
| 11 | -110.161 .64515 1 0.422 .941969 2.77638 2.91248 3.1142 |
| 12 | -110.083 .15683 1 0.692 .962357 2.79733 2.94477 3.1633 |
+-----+
* optimal lag
Endogenous: y
Exogenous: _cons

. dfuller y, lags(4)

Augmented Dickey-Fuller test for unit root

Variable: y                                Number of obs = 95
                                           Number of lags = 4

H0: Random walk without drift, d = 0

               Test               Dickey-Fuller
               statistic          critical value
               -----
               1%               5%               10%
-----
Z(t)          -3.622          -3.517          -2.894          -2.582
-----
MacKinnon approximate p-value for Z(t) = 0.0054.

.
.
.
. *(3) ACF and PACF
. //-----
.
.
. set scheme s2color

.
. ac y, name(acf_plot,replace) lags(20)
. pac y, name(pacf_plot, replace) lags(20)
. graph combine acf_plot pacf_plot, name(ACF_and_PACF_hw3, replace) ///
> title("ACF and PACF of y") xsize(8) ysize(4)

.
. graph export ACF_and_PACF_hw3.png, name(ACF_and_PACF_hw3) replace
file ACF_and_PACF_hw3.png saved as PNG format

.
. * Assumed models: ARMA(2,3) , ARMA(3,2), ARMA(3,3), ARMA(3,4), ARMA(4,3)
.
. qui arima y, ar(1/2) ma(1/3)

. estat ic

Akaike's information criterion and Bayesian information criterion

-----
Model |          N    ll(null)    ll(model)      df          AIC          BIC
-----+-----
. |          100          .   -124.3935        7    262.7871    281.0233
-----
Note: BIC uses N = number of observations. See [R] BIC note.

```

```
.
. qui arima y, ar(1/3) ma(1/2)
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	100	.	-125.8966	7	265.7933	284.0295

Note: BIC uses N = number of observations. See [R] BIC note.

```
.
. qui arima y, ar(1/3) ma(1/3)
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	100	.	-124.1635	8	264.327	285.1684

Note: BIC uses N = number of observations. See [R] BIC note.

```
.
. qui arima y, ar(1/3) ma(1/4)
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	100	.	-123.6252	9	265.2504	288.6969

Note: BIC uses N = number of observations. See [R] BIC note.

```
.
. qui arima y, ar(1/4) ma(1/3)
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	100	.	-123.5017	9	265.0035	288.45

Note: BIC uses N = number of observations. See [R] BIC note.

```
.
. * ARMA(2,3) has the lowest AIC and BIC
.
```

```

. *
. Estimation
. *****
.
. arima y, ar(1/2) ma(1/3)

```

```

(setting optimization to BHHH)
Iteration 0: log likelihood = -126.39398
Iteration 1: log likelihood = -124.96963
Iteration 2: log likelihood = -124.54717
Iteration 3: log likelihood = -124.46603
Iteration 4: log likelihood = -124.4234
(switching optimization to BFGS)
Iteration 5: log likelihood = -124.40873
Iteration 6: log likelihood = -124.3947
Iteration 7: log likelihood = -124.39371
Iteration 8: log likelihood = -124.39355
Iteration 9: log likelihood = -124.39355

```

ARIMA regression

```

Sample: 1 thru 100
Log likelihood = -124.3935
Number of obs      =      100
Wald chi2(5)       =      867.40
Prob > chi2        =      0.0000

```

		OPG		z	P> z	[95% conf. interval]	
	y	Coefficient	std. err.				
y							
	_cons	.2189163	.5099523	0.43	0.668	-.7805718	1.218404
ARMA							
	ar						
	L1.	1.601061	.1463742	10.94	0.000	1.314172	1.887949
	L2.	-.7312467	.1195333	-6.12	0.000	-.9655276	-.4969657
	ma						
	L1.	-.3519961	.1448399	-2.43	0.015	-.6358771	-.0681152
	L2.	-.3742514	.1267091	-2.95	0.003	-.6225967	-.1259061
	L3.	.4758397	.1156432	4.11	0.000	.2491833	.7024962
	/sigma	.8278788	.0570236	14.52	0.000	.7161147	.9396429

Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.

```

. estat ic

```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	100	.	-124.3935	7	262.7871	281.0233

Note: BIC uses N = number of observations. See [R] BIC note.

```

. estat aroots

```

Eigenvalue stability condition

Eigenvalue	Modulus
.8005303 + .3006624i	.85513
.8005303 - .3006624i	.85513

All the eigenvalues lie inside the unit circle.
AR parameters satisfy stability condition.

Eigenvalue stability condition

Eigenvalue		Modulus
-	.8175946	.817595
.	.5847954 + .4899123i	.762889
.	.5847954 - .4899123i	.762889

All the eigenvalues lie inside the unit circle.
MA parameters satisfy invertibility condition.

```
.
.
. *
. Diagnostic Test
. *****
.
.
. predict res , residuals

.
. ac res, name(res_acf, replace) lags(20)
. pac res , name(res_pacf, replace) lags(20)
. graph combine res_acf res_pacf , xsize(8) ysize(4) title("ACF and PACF of Residual")
> ///
> name(res_acf_pacf_hw3, replace)

.
. graph export res_acf_pacf_hw3.png, name(res_acf_pacf_hw3) replace
file res_acf_pacf_hw3.png saved as PNG format

.
. wntestq res

Portmanteau test for white noise
-----
Portmanteau (Q) statistic = 33.0943
Prob > chi2(40) = 0.7721

.
.
. *
. 10 Days Ahead Forecast
. *****
.
.
. keep if time <= 100
(0 observations deleted)

. qui arima y, ar(1/2) ma(1/3)
. tsappend, add(10)

.
. capture drop y_forecast y_mse y_se ci_lower ci_upper
. predict y_forecast, y dynamic(101)
. predict y_mse, mse dynamic(101)
```

```

. gen y_se = sqrt(y_mse)
. gen ci_lower = y_forecast - 1.96 * y_se
. gen ci_upper = y_forecast + 1.96 * y_se

. qui replace y_forecast = y if time == 100
. qui replace ci_lower = y if time == 100
. qui replace ci_upper = y if time == 100

. list time y_forecast ci_lower ci_upper if time >= 99 & time <= 103
+-----+
| time   y_forec~t   ci_lower   ci_upper |
+-----+
99. |    99   -.1943956   -1.817038    1.428247 |
100. |   100   -1.854962   -1.854962   -1.854962 |
101. |   101   -1.146127   -2.768769    .4765154 |
102. |   102   -1.122645   -3.718955    1.473665 |
103. |   103   -1.124413   -4.09876    1.849934 |
+-----+

.
.
. set scheme white_tableau

. twoway (rarea ci_lower ci_upper time if time >= 100, color("gs12") lwidth(medthick))
> ///
> (line y time if time <= 100 & time >= 1, lcolor("33 37 41") lwidth(medthin))
> ///
> (line y_forecast time if time >= 100, lcolor("255 68 69") lpattern(dash) lwid
> th(medthick)), ///
> xline(100, lcolor(gs10) lwidth(medthick) lpattern(shortdash)) ///
> graphregion(color(white)) ///
> title("{bf:ARMA(2,3): Historical Data vs. Dynamic Forecast}", color("108 117
> 125")) ///
> xtitle("Time Period", color("108 117 125")) ///
> ytitle("Value", color("108 117 125")) ///
> legend(order(2 "Actual Data" 1 "95% CI" 3 "Forecast") pos(11) ring(0) ///
> cols(1) color("108 117 125") ) ///
> ylabel(, format(%9.0f) nogrid) ///
> xlabel(0(25)100 110, labsize(small)) ///
> xsize(6.5) ysize(4) name(forecast_plot,replace)

. graph export forecast_plot.png, name(forecast_plot) replace
(file forecast_plot.png not found)
file forecast_plot.png saved as PNG format

. log close
  name: <unnamed>
  log: F:\ECON 408\Practice do file\Homework-3.log
  log type: text
  closed on: 16 May 2026, 12:56:55
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> -----

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